

# RunPod Serverless Endpoint

FLUX.1-dev text-to-image, deployed and measured

|              |                                                                                                                             |
|--------------|-----------------------------------------------------------------------------------------------------------------------------|
| Submitted by | Jimmie Young                                                                                                                |
| Date         | 22 September 2026                                                                                                           |
| Repository   | <a href="https://github.com/jimmieyoung3/runpod-flux-serverless">https://github.com/jimmieyoung3/runpod-flux-serverless</a> |
| Endpoint ID  | ux61jghq0twkgq                                                                                                              |
| Image        | docker.io/jimmieyoung3/flux-runpod:v1 (private, 29.87 GB)                                                                   |
| Model        | black-forest-labs/FLUX.1-dev, bfloat16                                                                                      |

## Summary

A RunPod serverless endpoint that accepts a text prompt over HTTP and returns a generated image. The FLUX.1-dev weights are baked into the Docker image, so a worker loads the model from local disk in **7.2 seconds** rather than pulling 33.7 GB from the Hugging Face Hub. Warm generation of a 1024×1024 image at 28 steps takes **14.1 s on an A100** or **30.3 s on an A40**, at a cost of roughly **\$0.0103 per image** on either. Total cost of building, deploying and benchmarking the whole exercise was **\$0.85**.

The sections below cover the architecture, the handler and image design, two platform constraints that materially shaped the build, the measured results, and what I would change for production.



Endpoint output. Prompt: “a cinematic photograph of a glass terrarium on a windowsill at golden hour, shallow depth of field, 85mm”. Seed 42, 1024×1024, 28 steps.

# 1. Architecture

The endpoint is a single-purpose GPU worker. RunPod owns the queue, autoscaling and HTTP surface; the worker owns model loading and inference.

```
client  ■■ POST /runsync {"input":{"prompt":...}} ■■■■ RunPod queue
          ■
          ▼
GPU worker (0-2, scale to zero)
  ■■ import: FluxPipeline.from_pretrained(/models/flux)
  ■■ warmup: one 512px / 1-step pass
  ■■ per request: validate ■ generate ■ encode base64
```

## Three decisions define the design.

| Decision                                      | Rationale                                                                     | Measured effect                                     |
|-----------------------------------------------|-------------------------------------------------------------------------------|-----------------------------------------------------|
| Weights baked into the image                  | Removes the Hub from the critical path; a cold worker reads from local disk.  | Model ready in <b>7.2 s</b> vs ~10 min from the Hub |
| Pipeline loaded at module import              | RunPod attributes import time to worker start-up, not to billed request time. | Warm requests bill only inference                   |
| One job per worker (concurrency_modifier = 1) | Generation saturates the GPU; two concurrent 1024px jobs slow each other.     | Execution stayed flat at 30.4 s under 2× load       |

A one-step warm-up pass runs at start-up so the first paying request does not absorb CUDA kernel autotuning and lazy initialisation.

# 2. The handler

src/handler.py is the RunPod entry point; src/predict.py owns the pipeline and src/schema.py validates input. Validation is deliberately free of torch and diffusers so it unit-tests on any machine — 25 tests run without a GPU.

```
def handler(job):
    params = validate(job.get("input"), DEFAULTS)
    result = predictor.generate(params)
    return {"images": [...], "seed": result.seed, "metrics": {...}}

runpod.serverless.start({"handler": handler,
                        "concurrency_modifier": adjust_concurrency})
```

## Notable details

| Behaviour                                    | Why                                                                                                                                                                                    |
|----------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Dimensions snap to a multiple of 16          | FLUX packs latents 2×2 on top of the VAE's 8× downscale. Other values silently produce a differently sized image, so the handler rounds and reports rather than failing.               |
| No negative_prompt                           | FLUX.1 is guidance-distilled and FluxPipeline has no negative conditioning. Silently accepting the field would be worse than rejecting it.                                             |
| Errors return HTTP 200 with an error field   | RunPod reserves non-2xx for platform failures; application errors belong in the body. Validation errors are separated from inference errors so a client mistake does not burn a retry. |
| Booleans rejected where numbers are expected | bool subclasses int in Python, so steps=True would otherwise silently mean 1 step.                                                                                                     |

Optional S3 delivery

Base64 keeps the endpoint dependency-free, but exceeds RunPod's ~20 MB response ceiling at about four 1024px PNGs. Setting `BUCKET_ENDPOINT_URL` switches to presigned URLs.

## 3. Two platform constraints

The image is 29.87 GB, which makes *where you build it* an engineering problem. Both constraints below were discovered during the build and are documented in the repository.

### Buildah and Podman cannot run inside a RunPod pod

RunPod pods have no `CAP_SYS_ADMIN`, and the hosts set `apparmor_restrict_unprivileged_userns=1`, so `unshare(CLONE_NEWUSER)` is denied. Both tools re-exec into a user namespace at start-up — before they honour `--isolation chroot` — so even `buildah containers` fails. Kaniko uses no namespaces and is the workable builder. RunPod's own in-pod build tutorial reaches for Bazel and crane for the same reason.

### Kaniko destroys the pod it runs in

Kaniko extracts the base image over `/`, deleting the userland that `sshd` depends on. SSH dies mid-build and does not recover; preserving the `sshd` binary via `--ignore-path` is not enough, because its PAM and `libc` dependencies go too. The consequence is architectural: Kaniko must **push the image itself**, because building to a local tarball strands the result on a pod you can no longer reach. The build runs detached under `setsid` and is monitored from outside against the registry.

### Keeping the token out of the image

Kaniko does not implement `RUN --mount=type=secret`. Passing the Hugging Face token as a build argument would persist it in the image history, so instead the weights are downloaded into the build context beforehand and simply copied in. The token never enters the image, its history or its layers — verified by auditing the published image config, which contains zero token-shaped strings. The cost is peak build disk, since the weights exist twice during the build.

RunPod's GitHub integration would otherwise be the natural choice. It is ruled out on two counts: it caps `docker build` at 30 minutes, and it exposes no build-time secret, which a gated repository such as FLUX.1-dev requires.

### Build pipeline as deployed

| Stage                               | Where                              | Time          |
|-------------------------------------|------------------------------------|---------------|
| Fetch 33.7 GB of weights (79 files) | RunPod CPU pod, 32 vCPU            | ~3 min        |
| Kaniko build + push of 29.87 GB     | same pod, detached                 | ~26 min       |
| Total build cost                    | two pods, incl. one failed attempt | <b>\$0.66</b> |

Root-level single-file checkpoints in the FLUX.1-dev repository duplicate the sharded `diffusers` folders that `FluxPipeline` actually loads. Excluding them cut the download from 57.8 GB to 33.7 GB.

## 4. Results

Measured against the live endpoint on 22 September 2026. `executionTime` is what RunPod bills; `delayTime` is queue plus worker start-up.

### Latency — 1024×1024, 28 steps, guidance 3.5

| Scenario                                     | delayTime     | executionTime                   | Wall   |
|----------------------------------------------|---------------|---------------------------------|--------|
| First-ever cold start (image on no host yet) | <b>609 s</b>  | 0.04 s                          | 650 s  |
| Cold start, image cached on host (A100)      | <b>15.5 s</b> | 16.0 s                          | 35.2 s |
| Warm, A40 — mean of 5                        | 0.69 s        | <b>30.32 s</b> ( $\sigma$ 0.06) | 33.1 s |
| Warm, A100 — mean of 4                       | 0.80 s        | <b>14.07 s</b> ( $\sigma$ 0.04) | 16.0 s |
| 2 concurrent, A40                            | 16.42 s       | 30.40 s                         | 49.9 s |

A standard deviation of 0.06 s across warm runs confirms the pipeline stays resident and nothing is re-loaded per request. Model load from the baked weights is 7.2–8.0 s.

### Cost

| GPU             | \$/hr  | s/image | \$/image        | Images per \$1 |
|-----------------|--------|---------|-----------------|----------------|
| A40 48 GB       | \$1.22 | 30.32   | <b>\$0.0103</b> | 97             |
| A100-SXM4 80 GB | \$2.72 | 14.07   | <b>\$0.0106</b> | 93             |

The most useful finding: **cost per image is nearly GPU-independent**. A 2.2× faster card costs 2.2× more per second, so the two differ by 3%. Choose the GPU tier for the latency you need, not to save money — what actually moves cost is step count, resolution, and idle workers.

### What a 30 GB image costs you

The first request sat in `throttled` for roughly 10 minutes before a worker was placed, and widening from 7 to 11 GPU tiers did not help — which points at the image rather than GPU scarcity. A host must cache ~30 GB before it can start a worker, a smaller pool than “any host with a free GPU”. So the baked-weights decision has a measurable cost as well as a measurable benefit:

|                        | Baked weights (deployed) | Network volume (fallback)            |
|------------------------|--------------------------|--------------------------------------|
| Model available in     | <b>7.2 s</b>             | ~10 min first boot, then volume read |
| First scheduling       | ~10 min throttle         | fast — ~9 GB image                   |
| Runtime Hub dependency | none                     | first boot only                      |

For a steady endpoint the baked image is the better trade. For one that scales from zero often, or across many regions, the volume variant schedules faster — it is implemented in the repository as `Dockerfile.volume`.

## 5. Sample generations

Unmodified endpoint output. All 28 steps, guidance 3.5.



Seed 7 — “a hand-lettered enamel shop sign reading FLUX ON RUNPOD...”. Text renders correctly.



Seed 1234 — “an isometric cutaway of a tiny mechanical workshop... tilt-shift”.



Seed 2026 — “a lone lighthouse on a basalt cliff during a winter storm”. This was the cold-start request.



Seed 99, 832×1216 — non-square output exercising the dimension snapping in schema.py.

## 6. Using the endpoint

```
curl -X POST https://api.runpod.ai/v2/ux61jghq0twkgq/runsync \
  -H "Authorization: Bearer $RUNPOD_API_KEY" \
  -H "Content-Type: application/json" \
  -d '{"input": {"prompt": "a red fox in a snowy forest", "seed": 7}}'
```

Or with the repository's client, which saves the decoded image and prints the metrics:

```
python client/call_endpoint.py "a red fox in a snowy forest" --seed 7
python client/call_endpoint.py --health # model + GPU report
python client/benchmark.py --runs 5 --gpu-rate 0.000339
```

### Request fields

| Field               | Default  | Notes                                              |
|---------------------|----------|----------------------------------------------------|
| prompt              | required | ≤ 2000 characters                                  |
| width, height       | 1024     | snapped to a multiple of 16, clamped 256–1536      |
| num_inference_steps | 28       | 1–50                                               |
| guidance_scale      | 3.5      | 0–20                                               |
| seed                | random   | the seed used is always returned                   |
| num_images          | 1        | 1–4                                                |
| output_format       | PNG      | PNG, JPEG or WEBP                                  |
| action              | —        | "health" returns model/GPU info without generating |

## 7. What I would change for production

| Change                                         | Why                                                                                                                                                                 |
|------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Ship the network-volume variant across regions | The 30 GB image limits which hosts can place a worker. A ~9 GB image plus a pre-populated volume schedules faster and still avoids a per-boot download.             |
| Return presigned S3 URLs, not base64           | Base64 is convenient for review but inflates payloads by a third and hits RunPod's ~20 MB response ceiling at about four 1024px PNGs. The code path already exists. |
| Pin the GPU tier once traffic is known         | Latency varied 14–30 s purely on which card was scheduled. Latency-sensitive traffic should pin the faster tier; batch traffic should not pay for it.               |
| Add request-level observability                | The handler returns per-request metrics, but nothing aggregates them. Prometheus or a log sink would make regressions visible.                                      |
| Quantised variants for throughput              | fp8 or NF4 transformer weights roughly halve both load time and VRAM, at some quality cost — worth measuring against the bf16 baseline recorded here.               |

Full source, tests and deployment runbook: <https://github.com/jimmieyoung3/runpod-flux-serverless>

FLUX.1-dev weights are covered by the FLUX.1-dev Non-Commercial Licence, which is why the built image is held in a private registry and not redistributed.